

Evolution of Knowledge Areas in Publications

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Motivation

Understanding how research communities evolve in citation networks is essential for identifying emerging trends and structural changes.

However, distinguishing real structural evolution from noise remains challenging.

We propose a framework for tracking community evolution over time using graph representation learning, clustering, and temporal analysis.

Challenges and Solutions

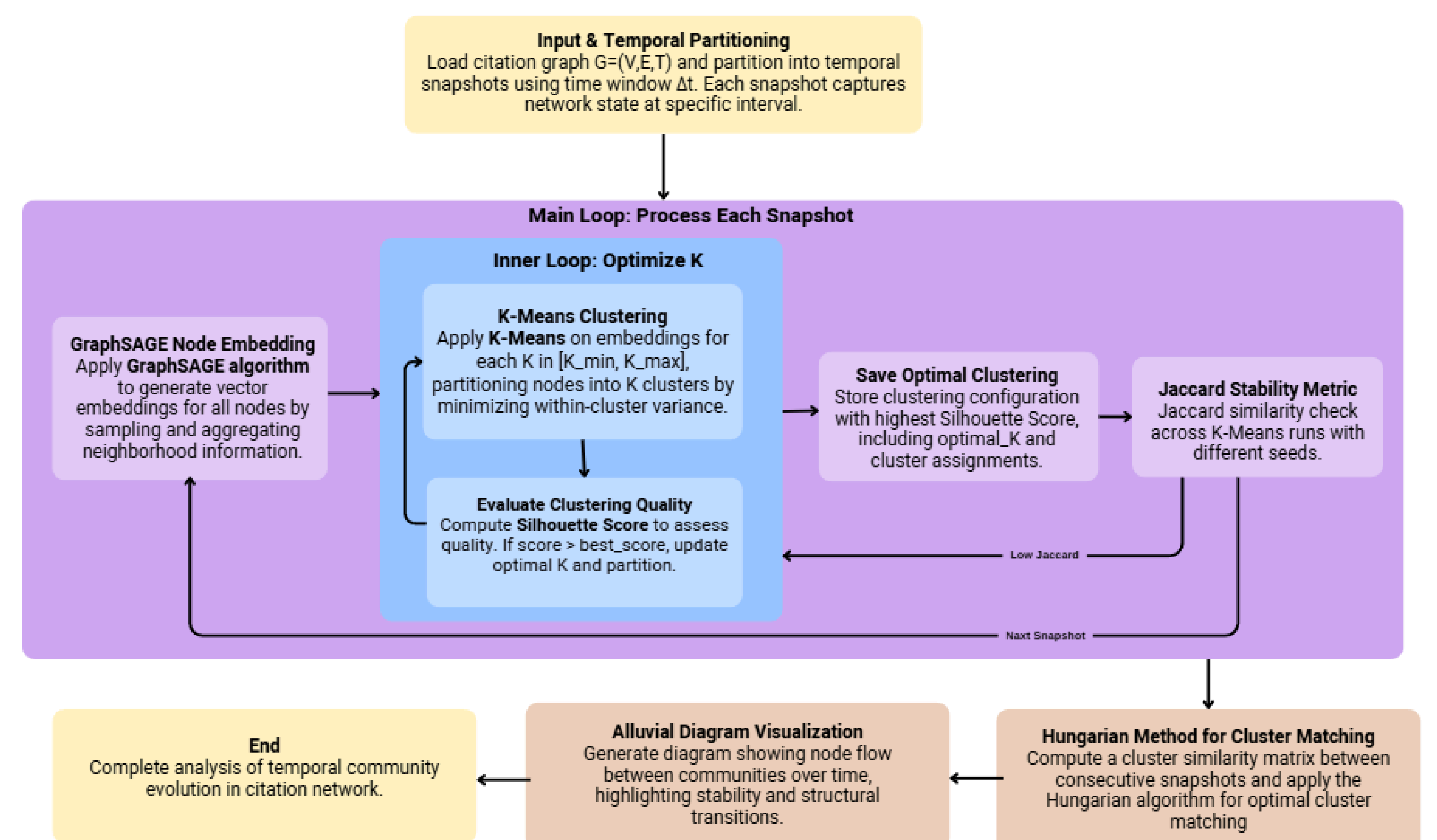
- Noise & clustering instability → mitigated using GraphSAGE + repeated K-Means
- Temporal inconsistency → alignment via Hungarian algorithm + Jaccard similarity
- Cluster selection → optimized using Silhouette Score
- Scalability → handled via snapshot-based processing
- Interpretation complexity → addressed using Alluvial visualization

Proposed Solution

Pipeline:

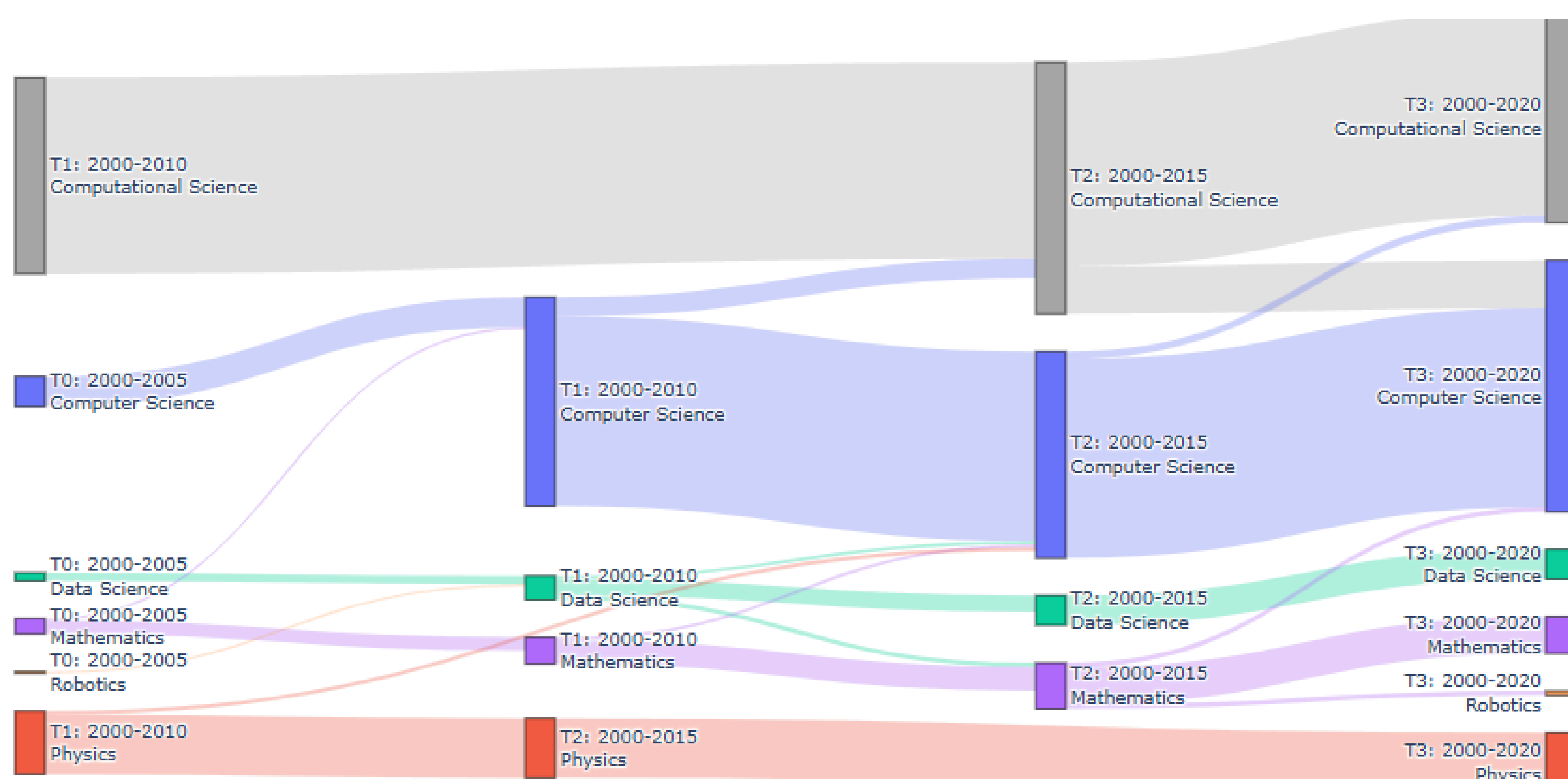
- Graph construction from citation data
- Temporal snapshot generation
- Node embeddings using GraphSAGE
- Clustering via K-Means (k selected using Silhouette Score)
- Temporal alignment using Hungarian algorithm + Jaccard similarity
- Visualization using Alluvial diagrams

DYNAMIC CITATION NETWORK ANALYSIS ALGORITHM



Results

Time Window	Papers	Edges	Categories	Best K	Silhouette Score	Robustness Mean	Robustness Std
2000–2005	2,466	706	7	10	0.896997	0.931288	0.068712
2000–2010	21,983	27,519	8	10	0.898320	0.908003	0.071905
2000–2015	25,502	30,917	8	9	0.822343	0.798616	0.079437
2000–2020	25,562	31,003	8	10	0.801391	0.854567	0.054467



Key Findings

- Computer Science forms a dominant and stable backbone across all time windows
- Computational Science emerges as a major community from 2010 onward
- Community similarity increases over time, indicating structural stabilization
- Smaller fields (AI, Robotics) remain more dynamic and fragmented

Conclusion

This work presents a scalable framework for analyzing temporal evolution in citation networks using graph representation learning and clustering.

The results reveal that citation networks evolve toward increasing structural stability, with dominant research domains acting as persistent cores.

The framework enables effective identification of emerging trends and community dynamics in large-scale, noisy data.

Software & Technologies

- Python, PyTorch, PyTorch Geometric, scikit-learn, pandas, NumPy
- Visualization: Matplotlib, Seaborn, Plotly
- Environment: Google Colab